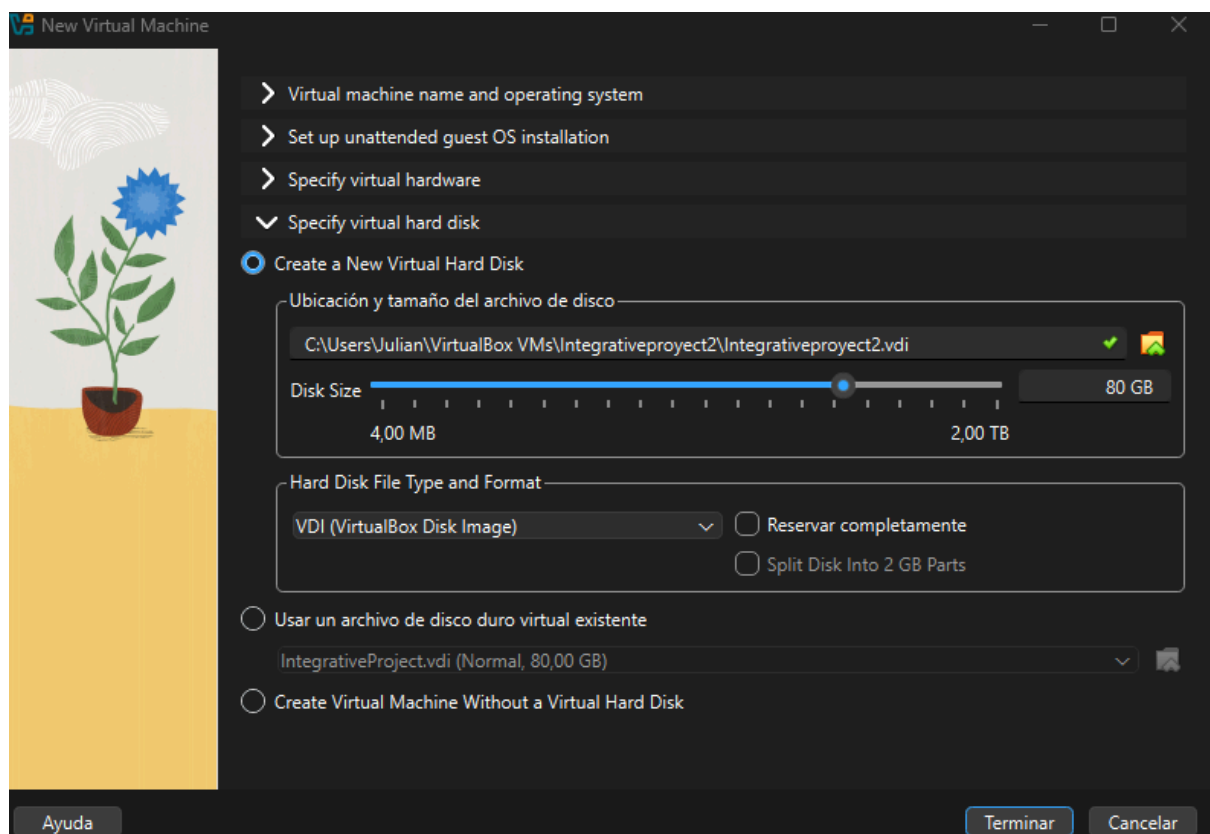
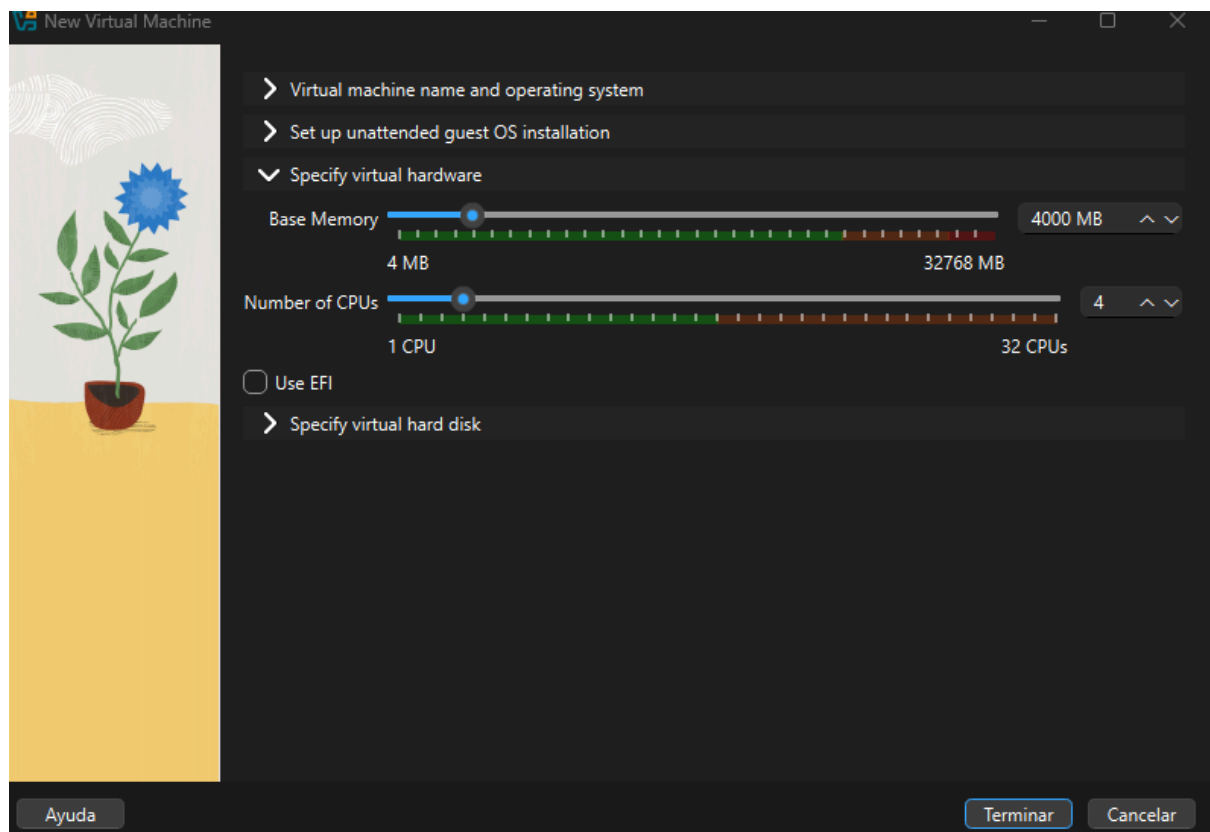




`sudo apt-add-repository universe` — habilita el repositorio "universe" de Ubuntu (paquetes mantenidos por la comunidad) para que `apt` pueda instalar software desde ahí.

`sudo apt-add-repository ppa:cubic-wizard/release` — agrega un repositorio externo (PPA) que contiene el paquete `cubic`, ya que este no está en los repositorios oficiales de Ubuntu.

`sudo apt update` — actualiza la lista de paquetes disponibles, descargando los índices más recientes de todos los repositorios configurados, incluyendo los dos que acabas de agregar.

`sudo apt install --no-install-recommends cubic` — instala el programa `cubic`, evitando instalar paquetes adicionales "recomendados" que no son estrictamente necesarios, para mantener la instalación más liviana.

```
ubuntu@ubuntu:~$ sudo apt-add-repository universe
Adding component(s) 'universe' to all repositories.
Press [ENTER] to continue or Ctrl-C to cancel.
Ign:1 file:/cdrom/resolute InRelease
Get:2 file:/cdrom/resolute Release [664 B]
Get:2 file:/cdrom/resolute Release [664 B]
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:5 http://archive.ubuntu.com/ubuntu resolute InRelease
Hit:6 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:7 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Reading package lists... Done
N: Skipping acquire of configured file 'restricted/binary-i386/Packages' as repository 'file:/cdrom/resolute InRelease' doesn't support architecture 'i386'
N: Skipping acquire of configured file 'main/binary-i386/Packages' as repository 'file:/cdrom/resolute InRelease' doesn't support architecture 'i386'
ubuntu@ubuntu:~$ sudo apt-add-repository ppa^C
ubuntu@ubuntu:~$ sudo apt-add-repository ppa:cubic-
ERROR: ppa 'cubic-/ppa' not found (use --login if private)
ubuntu@ubuntu:~$ sudo apt-add-repository ppa:cubic-wizard/release
```

```
ubuntu@ubuntu:~$ sudo apt-add-repository ppa:cubic-wizard/release
Repository: 'Types: deb
URIs: https://ppa.launchpadcontent.net/cubic-wizard/release/ubuntu/
Suites: resolute
Components: main
'
```

```
Description:
PPA for Cubic release branch
```



Custom Ubuntu ISO Creator

Cubic (Custom Ubuntu ISO Creator) is a GUI wizard to create a customized Ubuntu or Debian Live ISO image.

Website: <https://github.com/PJ-Singh-001/Cubic/wiki>

More info: <https://launchpad.net/~cubic-wizard/+archive/ubuntu/release>

Adding repository.

```
ubuntu@ubuntu:~$ sudo apt update
```

```
Ign:1 file:/cdrom resolute InRelease
```

```
Get:2 file:/cdrom resolute Release [664 B]
```

```
Get:2 file:/cdrom resolute Release [664 B]
```

```
Hit:4 http://archive.ubuntu.com/ubuntu resolute InRelease
```

```
Hit:5 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
```

```
Hit:6 http://security.ubuntu.com/ubuntu resolute-security InRelease
```

```
Hit:7 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
```

```
Hit:8 https://ppa.launchpadcontent.net/cubic-wizard/release/ubuntu resolute InRelease
```

206 packages can be upgraded. Run 'apt list --upgradable' to see them.

Notice: Skipping acquire of configured file 'restricted/binary-i386/Packages' as repository 'file:/cdrom resolute InRelease' doesn't support architecture 'i386'

Notice: Skipping acquire of configured file 'main/binary-i386/Packages' as repository 'file:/cdrom resolute InRelease' doesn't support architecture 'i386'

```
ubuntu@ubuntu:~$ sudo apt install -no-install-recommends cubic
```

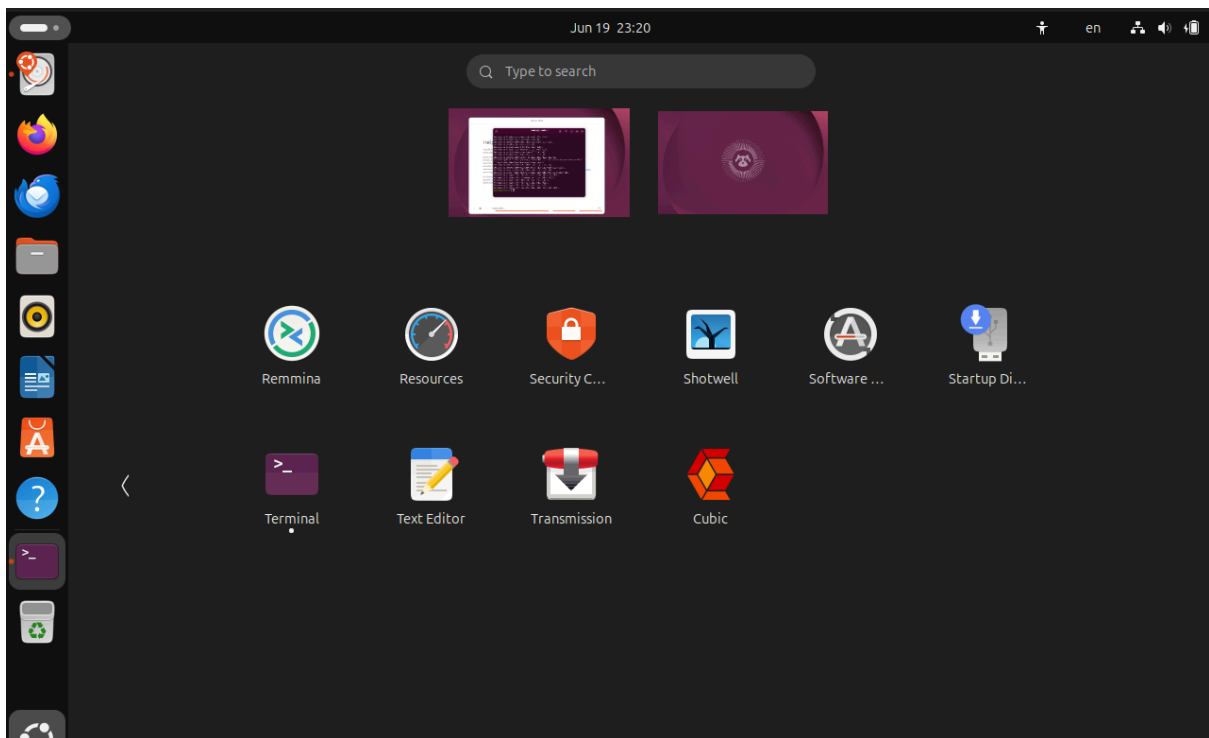
E: Command line option 'n' [from -no-install-recommends] is not understood in combination with the other options.

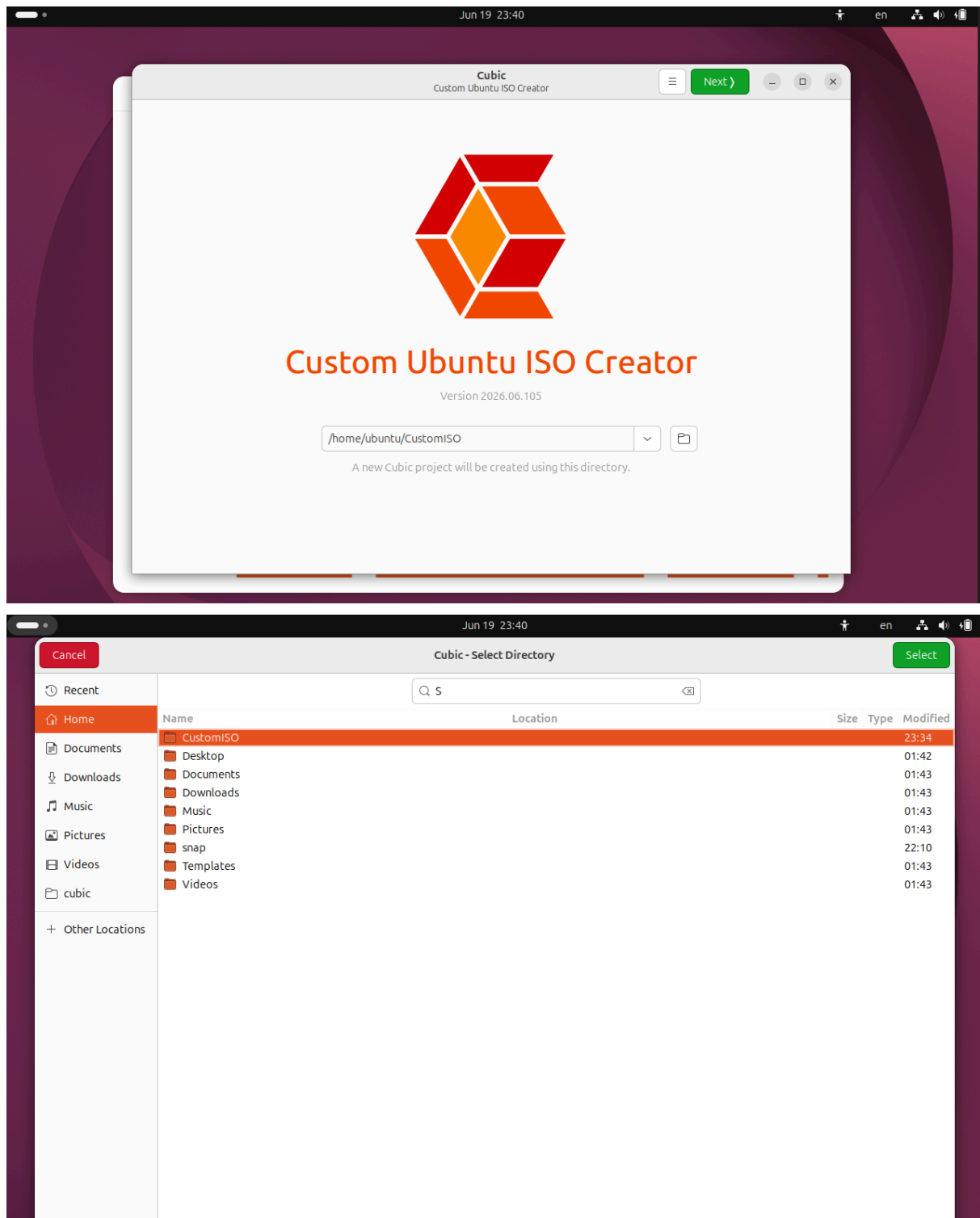
```
ubuntu@ubuntu:~$ sudo apt install --no-install-recommends cubic
```

```
ubuntu@ubuntu:~$ sudo apt install --no-install-recommends cubic
Installing:
  cubic
```

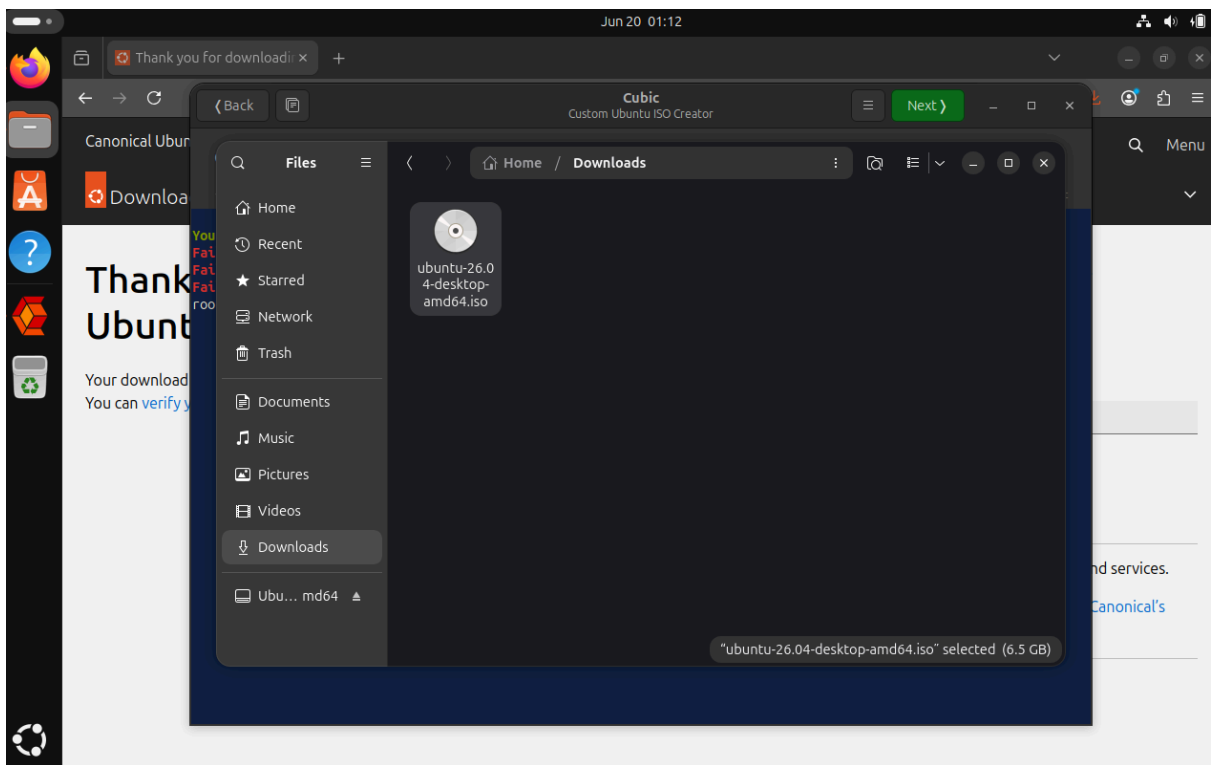
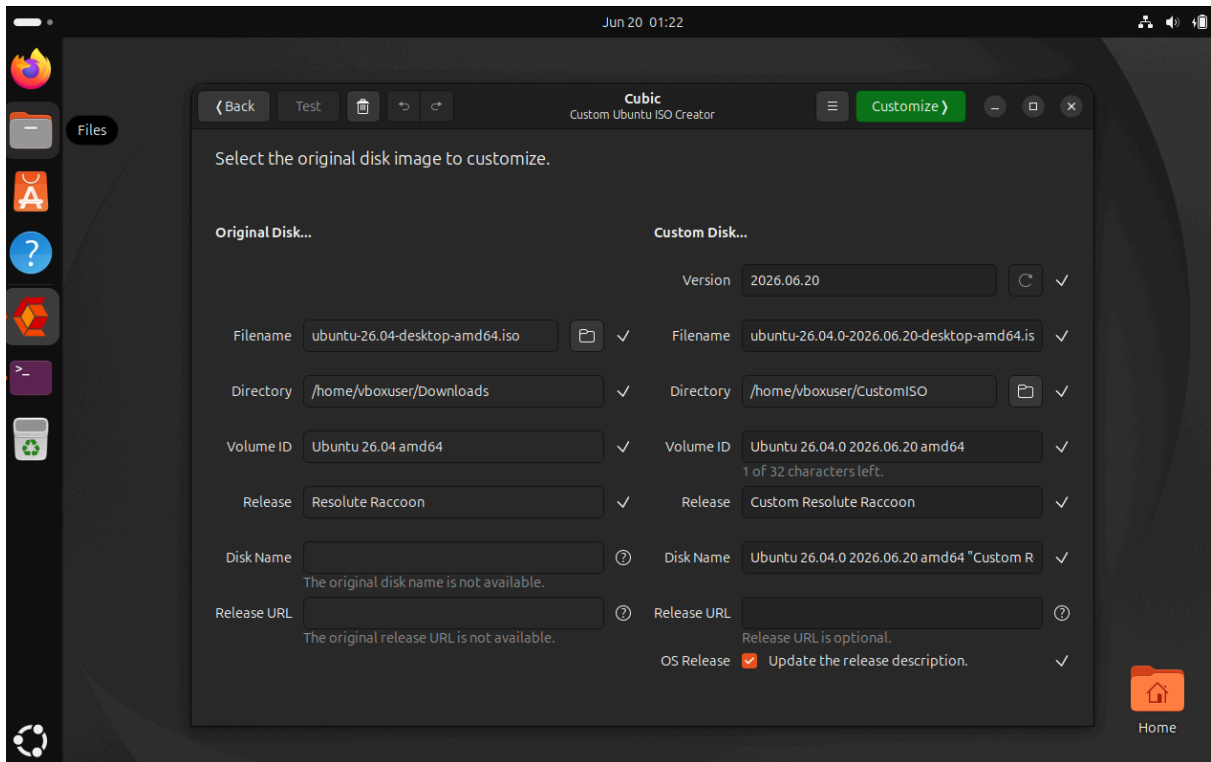
Installing dependencies:

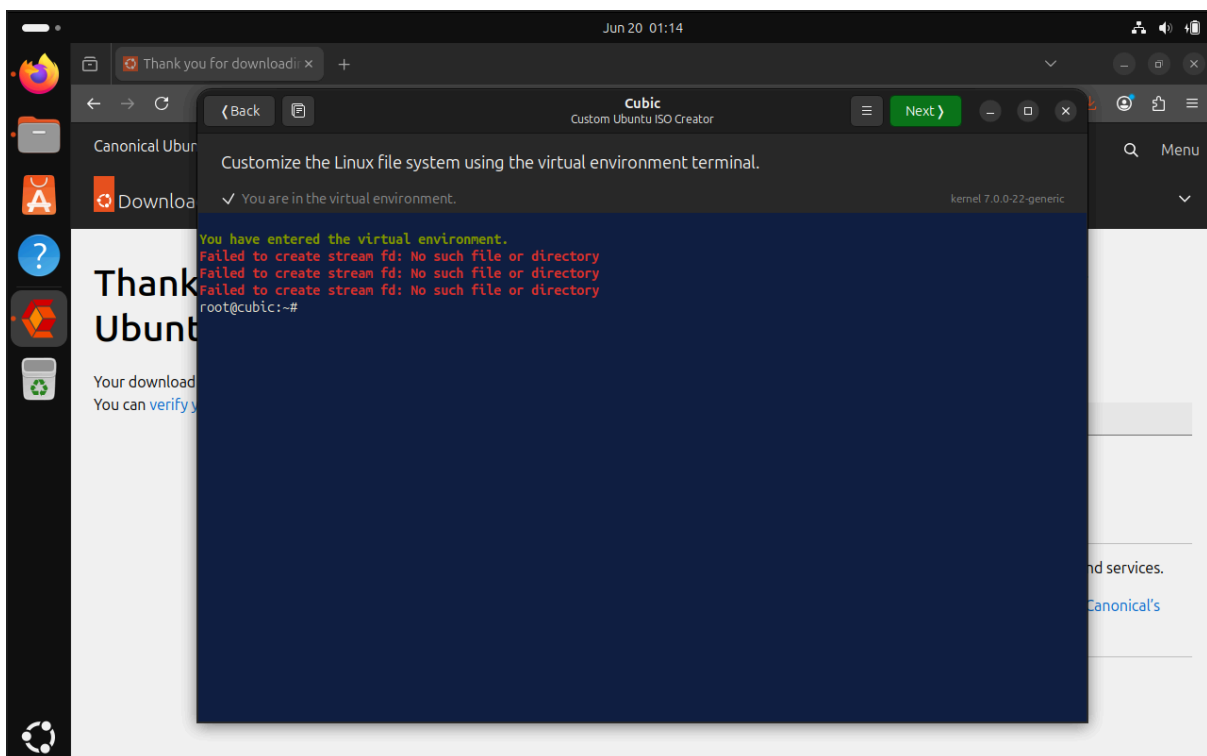
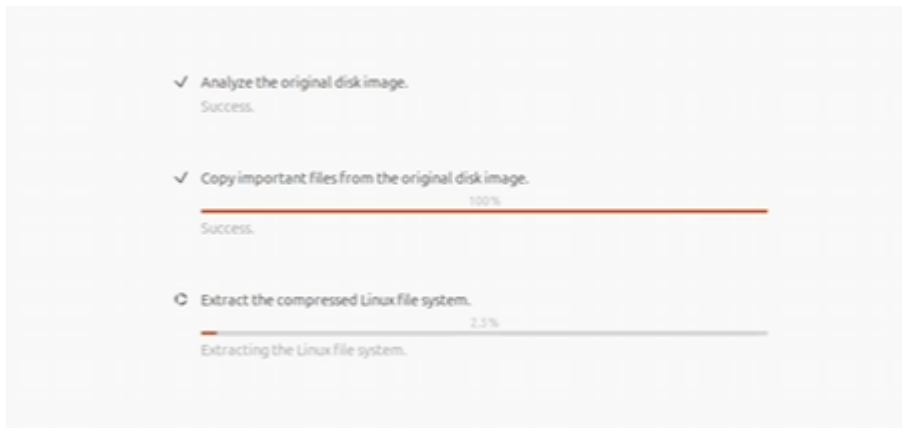
binwalk	libusbredirparser1t64
genisoimage	libvirglrenderer1
gir1.2-gtksource-4	libxss1
ipxe-qemu	python3-argcomplete
isolinux	python3-binwalk
libcacard0	python3-icu
libdaxctl1	python3-magic
libfdt1	python3-packaging
libgtksourceview-4-0	python3-pyasyncore
libgtksourceview-4-common	python3-pydbus
libjack-jackd2-0	python3-pyinotify
libmpathcmd0	qemu-system-common
libmpathpersist0	qemu-system-data
libmultipath0	qemu-system-gui
libndctl6	qemu-system-modules-opengl





Cubic pide una ISO para configurar, se re instala el iso en la MV y se espera a la descarga de archivos





replacing firefox with librewolf

apt remove -y firefox

apt update

apt install -y extrepo

extrepo enable librewolf

apt update

apt install -y librewolf

```
Cubic
Custom Ubuntu ISO Creator

Customize the Linux file system using the virtual environment terminal.

✓ You are in the virtual environment. kernel 7.0.0-22-generic

You have entered the virtual environment.
Failed to create stream fd: No such file or directory
Failed to create stream fd: No such file or directory
Failed to create stream fd: No such file or directory
root@cubic:~# apt remove -y firefox
REMOVING:
  firefox

Summary:
  Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 0
  Freed space: 124 kB

(Reading database ... 208479 files and directories currently installed.)
Removing firefox (1:1snap1-0ubuntu8) ...
Processing triggers for hicolor-icon-theme (0.18-2build1) ...
root@cubic:~# apt install -y wget gnupg
Error: Unable to locate package 0y
root@cubic:~# apt install -y wget gnupg
wget is already the newest version (1.25.0-2ubuntu4).
wget set to manually installed.
gnupg is already the newest version (2.4.8-4ubuntu3).
gnupg set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0
root@cubic:~#
```

Repo oficial de Neovim (PPA con la última versión estable)

```
sudo apt install -y software-properties-common
sudo add-apt-repository -y ppa:neovim-ppa/stable
sudo apt update
sudo apt install -y neovim git curl
```

```
root@cubic:~# add-apt-repository -y ppa:neovim-ppa/stable
PPA publishes dbgsym, you may need to include 'main/debug' component
Repository: 'Types: deb
URIs: https://ppa.launchpadcontent.net/neovim-ppa/stable/ubuntu/
Suites: resolute
Components: main
'
Description:
  Neovim releases backported from Debian.

Homepage: https://neovim.io
Source/Issues: https://github.com/neovim/neovim
Documentation: https://neovim.io/doc
User Manual: https://neovim.io/doc/user
Neovim features: https://neovim.io/doc/user/nvim.html
Packaging: https://launchpad.net/neovim-ppa
More info: https://launchpad.net/~neovim-ppa/+archive/ubuntu/stable
Adding repository.
gpg: Warning: using insecure memory!
Ign:1 file:/cdrom/resolute InRelease
Err:2 file:/cdrom/resolute Release
  File not found - /cdrom/dists/resolute/Release (2: No such file or directory)
Hit:3 http://archive.ubuntu.com/ubuntu resolute InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
```

Crear una config básica de Neovim

```
mkdir -p /etc/skel/.config/nvim
cat > /etc/skel/.config/nvim/init.vim << 'EOF'
" Configuración básica de Neovim
set number
set relativenumber
set tabstop=4
```



```
set shiftwidth=4
set expandtab
set smartindent
syntax on
EOF
```

```
root@cubic:~# mkdir -p /etc/skel/.config/nvim
root@cubic:~# cat > /etc/skel/.config/nvim/init.vim << 'EOF'
> set number
> set relativenumber
> set tabstop=4
> set shiftwidth=4
> set expandtab
> set smartindent
> syntax on
> EOF
root@cubic:~# cat /etc/skel/.config/nvim/init.vim
set number
set relativenumber
set tabstop=4
set shiftwidth=4
set expandtab
set smartindent
syntax on
root@cubic:~#
```

La config de Neovim de arriba ya queda en /etc/skel, pero puedes agregar más:

Ejemplo: un .bashrc personalizado

```
cat >> /etc/skel/.bashrc << 'EOF'
```

```
> Alias personalizados de la distro
alias ll='ls -alh'
alias update='sudo apt update && sudo apt upgrade -y'
EOF
```

- **Ejemplo: carpeta de Documentos/Proyectos por defecto**

```
mkdir -p /etc/skel/Proyectos
```

```
root@cubic:~# mkdir -p /etc/skel/.config/nvim
root@cubic:~# cat > /etc/skel/.config/nvim/init.vim << 'EOF'
> set number
> set relativenumber
> set tabstop=4
> set shiftwidth=4
> set expandtab
> set smartindent
> syntax on
> EOF
root@cubic:~# cat /etc/skel/.config/nvim/init.vim
set number
set relativenumber
set tabstop=4
set shiftwidth=4
set expandtab
set smartindent
syntax on
root@cubic:~# cat >> /etc/skel/.bashrc << 'EOF'
> #Alias personalisda de la distro
> alias ll='ls -alh'
> alias update='apt update && apt upgrade -y'
> alias edit='nvim'
> EOF
root@cubic:~# mkdir -p /etc/skel/Proyectos
root@cubic:~#
```

Primero se instala el tema que sea (ejemplo: Yaru-dark o uno custom)

```
sudo apt install -y yaru-theme-gtk yaru-theme-icon yaru-theme-sound
```

Crear el override de gschema para que sea el DEFAULT del sistema

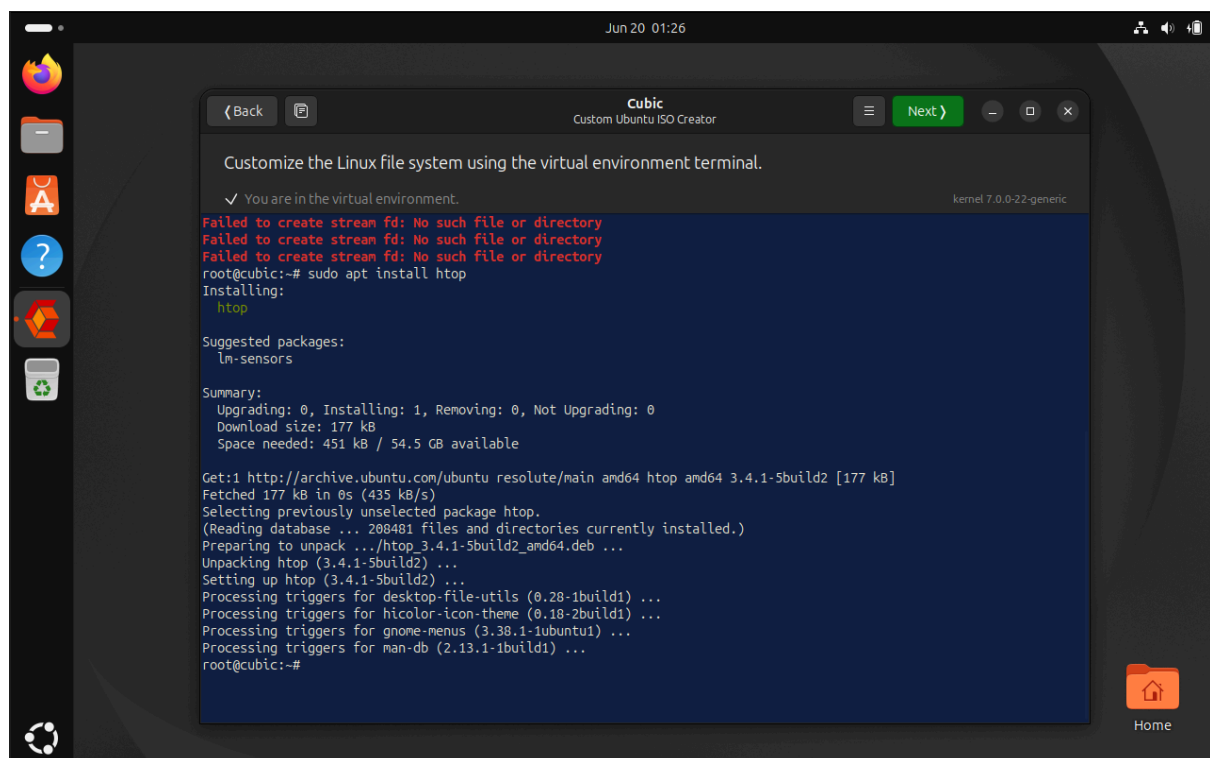
```
sudo bash -c 'cat > /usr/share/glib-2.0/schemas/zz_custom-theme.gschema.override << EOF
[org.gnome.desktop.interface]
gtk-theme="Yaru-dark"
icon-theme="Yaru"
color-scheme="prefer-dark"
EOF'
```

Compilar los schemas para que tome efecto

```
sudo glib-compile-schemas /usr/share/glib-2.0/schemas/
```

```
root@cubic:~# bash -c 'cat > /usr/share/glib-2.0/schemas/zz_custom-theme.gschema.override << EOF
> [org.gnome.desktop.interface]
> gtk-theme="Yaru-dark"
> icon-theme="Yaru"
> color-scheme="prefer-dark"
> EOF'
```

installing httpd



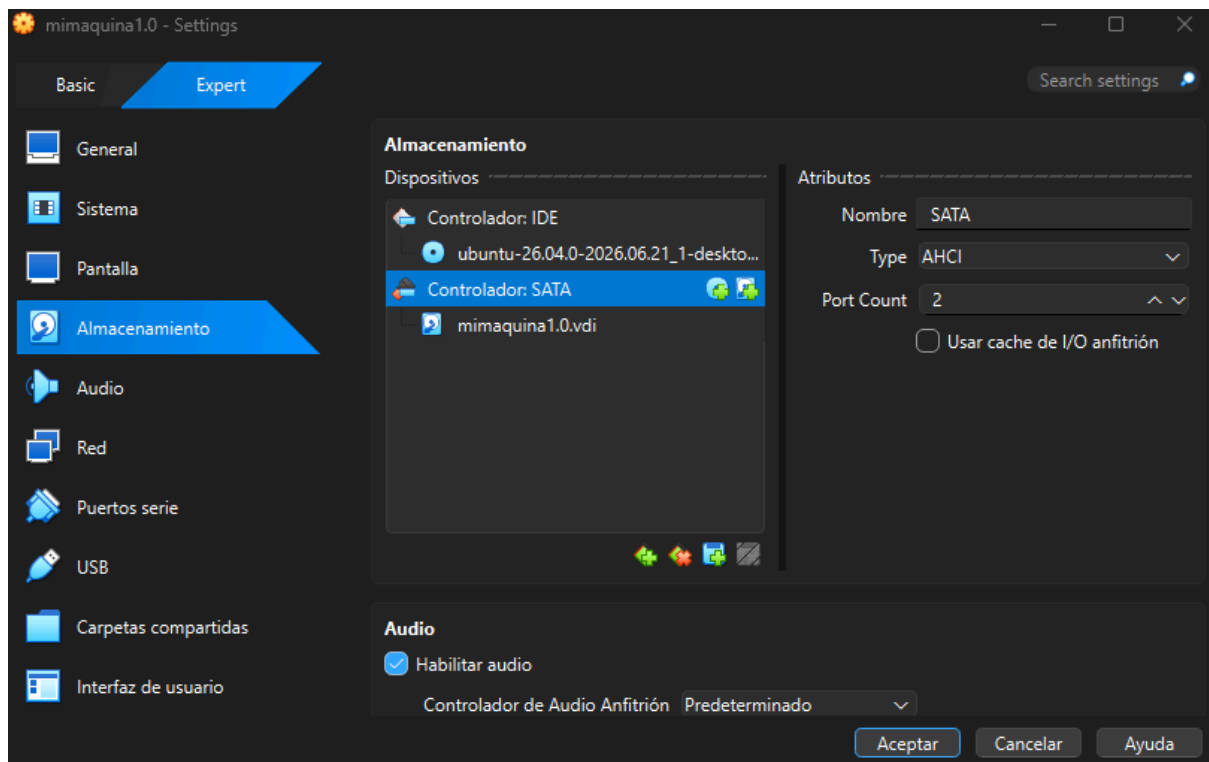
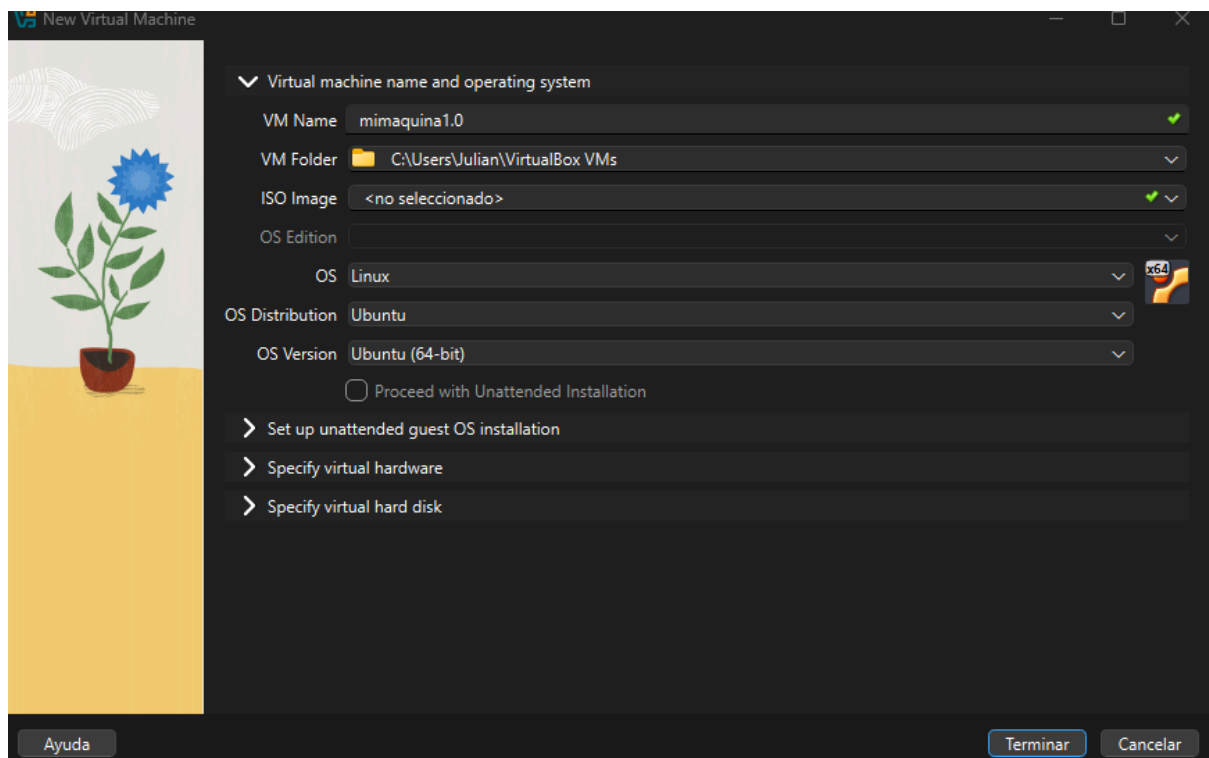
installing apache

```
root@cubic:~# apt update
Ign:1 file:/cdrom/resolute InRelease
Err:2 file:/cdrom/resolute Release
  File not found - /cdrom/dists/resolute/Release (2: No such file or directory)
Hit:3 http://archive.ubuntu.com/ubuntu/resolute InRelease
Hit:4 http://archive.ubuntu.com/ubuntu/resolute-updates InRelease
Hit:5 http://security.ubuntu.com/ubuntu/resolute-security InRelease
Hit:6 http://archive.ubuntu.com/ubuntu/resolute-backports InRelease
Ign:7 https://ppa.launchpadcontent.net/neovim-ppa/stable/ubuntu/resolute InRelease
Err:8 https://ppa.launchpadcontent.net/neovim-ppa/stable/ubuntu/resolute Release
  404 Not Found [IP: 2620:2d:4000:1009::1ed 443]
Get:9 https://repo.librewolf.net/librewolf InRelease [5649 B]
Error: The repository 'file:/cdrom/resolute Release' does not have a Release file.
Notice: Updating from such a repository can't be done securely, and is therefore disabled by default.
Notice: See apt-secure(8) manpage for repository creation and user configuration details.
Error: The repository 'https://ppa.launchpadcontent.net/neovim-ppa/stable/ubuntu/resolute Release' does not have a Release file.
Notice: Updating from such a repository can't be done securely, and is therefore disabled by default.
Notice: See apt-secure(8) manpage for repository creation and user configuration details.
Notice: Skipping acquire of configured file 'main/binary-i386/Packages' as repository 'https://repo.librewolf.net/librewolf InRelease' doesn't support architecture 'i386'
root@cubic:~# apt install -y apache2
Installing:
  apache2

Installing dependencies:
```

Press next and let the program load all the archives. When finished you will be left with an ISO

Se confirma la creacion de la iso booteable creando otra maquina virtual usando la iso como disco optico



```
==== Deployment Started ====
This process can take a few minutes to complete.
You may run "tail -f /var/log/lab-install.log" from another terminal session to see the progress of the deployment.
Deploying the lab...
OK: all containers appear to be running. Performing a couple of post provisioning steps...
OK: Lab is up and provisioned.
@Nailuj-glitch →/workspaces/kernel_integrativeProject/Black-Hat-Bash/lab (master) $
```

```
● @Nailuj-glitch →/workspaces/kernel_integrativeProject/Black-Hat-Bash/lab (master) $ sudo make test
Lab is up.
```

```
● @Nailuj-glitch →/workspaces/kernel_integrativeProject/Black-Hat-Bash/lab (master) $
```

```
● @Nailuj-glitch →/workspaces/kernel_integrativeProject/Black-Hat-Bash/lab (master) $ sudo docker ps --format "{{.Names}}"
p-web-02
p-jumpbox-01
p-web-01
p-ftp-01
c-backup-01
c-redis-01
c-db-02
c-db-01
```

```
● @Nailuj-glitch →/workspaces/kernel_integrativeProject/Black-Hat-Bash/lab (master) $ ip addr | grep "br_"
114: br_corporate: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    inet 10.1.0.1/24 brd 10.1.0.255 scope global br_corporate
115: br_public: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    inet 172.16.10.1/24 brd 172.16.10.255 scope global br_public
116: veth781edc8@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br_corporate state UP group default
117: vethc582f18@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br_public state UP group default
118: veth45ed4c8@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br_corporate state UP group default
119: vetha18ad8d@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br_public state UP group default
120: veth08f6275@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br_corporate state UP group default
121: veth5ea72a4@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br_corporate state UP group default
122: veth633caf6@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br_corporate state UP group default
123: veth027e5ac@if3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br_public state UP group default
124: veth2392091@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br_corporate state UP group default
125: vethdfc7891@if3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br_public state UP group default
```

```
● @Nailuj-glitch →/workspaces/kernel_integrativeProject/Black-Hat-Bash/lab (master) $ sudo docker exec -it p-web-01 bash
root@p-web-01:/app# whoami
root
root@p-web-01:/app# hostname
p-web-01.acme-infinity-servers.com
root@p-web-01:/app# exit
exit
```